

Non-Functional Requirements (NFRs):

.1 Reliability & Availability

Since the system is medical and handles emergency cases and critical imaging equipment (CT/MRI), high reliability is mandatory.

- **Availability:** The system must maintain 99.9% uptime (High Availability). Any downtime could result in delayed critical scans and impact patient outcomes.
- **Fault Tolerance:** In the event of a radiology equipment failure, the system must remain operational. It should automatically trigger notifications to affected patients and facilitate rescheduling without a total system collapse.

.2 Security & Data Privacy

- **Authentication & Authorization:** Implementation of JWT (JSON Web Tokens) to ensure that only authorized personnel (e.g., receptionists or doctors) can modify schedules.
- **Data Privacy:** Sensitive patient information, such as names and phone numbers, must be encrypted within the database.
- **Input Validation:** Strict validation must be enforced to prevent SQL Injection attacks, particularly in personal data fields like "Name" and "Phone Number".

3- Maintainability & Development Standards

- **Coding Standards:** Adherence to a unified coding style to ensure the codebase is readable and maintainable by any team member without needing to consult the original author.

- **Testability:** The system architecture must support Unit Testing, specifically for the "Automated Appointment Selection" logic, to ensure accuracy and prevent regressions during updates.
- **Loose Coupling:** The Radiology Module (SCH-RAD) must remain independent. If other system modules fail, the radiology scheduling system should continue to function autonomously.

4 -Scalability

Extensibility: The system must allow for the seamless addition of new imaging modalities (e.g., Ultrasound) by simply updating "Device Capacity" in the database, rather than rewriting the core scheduling logic.